

AKASH YADAV

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SUMMARY

Ph.D. researcher in scientific machine learning (SciML) and computational science, developing trustworthy AI methods for scientific discovery. Experienced in uncertainty-aware scientific foundation models, stochastic attention mechanisms, probabilistic low-rank adaptation techniques, model calibration, Bayesian optimization, and simulation-based modeling. Strong background in Python/PyTorch, HPC workflows, finite element modeling, and automated experiment pipelines. Interested in building agentic AI systems that combine coding agents, reusable skills, evaluation harnesses, and computational automation to accelerate scientific discovery.

RESEARCH INTERESTS

Agentic AI for Scientific Discovery, Scientific ML, Scientific Foundation Models, Uncertainty Quantification

EDUCATION

Doctor of Philosophy, Civil Engineering (GPA: 4.0/4.0) 08/2023 - Present
Thesis: Quantifying and Reducing Model Error via Stochastic Representations
University of Houston (UH) - Houston, USA

Master of Technology (Research), Civil Engineering (GPA: 8.3/10) 10/2020 - 06/2023
Thesis: SHM accounting for Thermal Variability and Damage using ABC
Indian Institute of Science (IISc) - Bangalore, India

Bachelor of Technology, Civil Engineering (First Division, GPA: 8.75/10) 07/2014 - 05/2018
Indian Institute of Technology (IIT) - Roorkee, India

SKILLS

Agentic AI	Codex/Claude Code-assisted workflows, Multi-agent & automated research workflows
ML/Stats	Transformers, LoRA, PINNs, CNNs, GANs, Gaussian Processes, MCMC methods
Programming	Python (PyTorch, NumPy, SciPy), BO (GPyTorch, BoTorch), MATLAB, Julia
HPC/CAE	Distributed training, SLURM, Git, Abaqus, LS-DYNA, FEniCS, Gmsh

RESEARCH EXPERIENCE

Graduate Research Assistant - UH (Advisor: Dr. Ruda Zhang) 08/2023 - Present

- Developed inference-time stochastic attention mechanism for transformer-based scientific foundation models, enabling calibrated predictive uncertainty without retraining pretrained backbones.
- Built automated calibration and evaluation harnesses for uncertainty-aware SciML models, integrating stochastic inference, calibration baselines, benchmark comparisons, and reproducible reporting.
- Developed Bayesian optimization workflows for stochastic model calibration, reducing data requirements and accelerating hyperparameter tuning by 40×.
- Developed stochastic reduced-order modeling frameworks for model-error correction in computational mechanics, improving uncertainty-aware prediction under limited data.
- Proposed stochastic subspace models via probabilistic principal component analysis and bootstrap resampling for model-form error characterization in computational mechanics.

Research Project - Rice University 08/2024 - 12/2024

- Implemented a GAN-augmented physics-informed neural network pipeline for UQ, including model training, physics-loss integration, and benchmark evaluation inspired by Yang et al. (2019).

Graduate Research Assistant - IISc (Advisor: Dr. Ananth Ramaswamy) 10/2020 - 06/2023

- Developed an Approximate Bayesian Computation framework for finite-element-based structural damage detection under thermal variability, and extended it to account for damage-induced nonlinearity.

AGENTIC AI/SCIENTIFIC AUTOMATION PROJECTS

- Developing Codex/Claude Code-assisted workflows for SciML experimentation, using reusable instructions/skills to automate experiment setup, code generation, testing, evaluation, and result summarization.
- Building evaluation harnesses around verifiable outputs, including experiment logs, benchmark tables, diagnostic plots, regression checks, and reproducible reports.
- Designing workflow templates for uncertainty-aware SciML experiments, including stochastic inference, calibration metrics, baseline comparisons, and automated result packaging.

SELECTED PUBLICATIONS

1. **Yadav, A.**, Adebisi, T.A., & Zhang, R. (2026). "Calibrating Scientific Foundation Models with Inference-Time Stochastic Attention." *arXiv*. [arXiv:2604.19530](#). under review.
2. **Yadav, A.** & Zhang, R. (2025). "Bayesian Optimization under Uncertainty for Training a Scale Parameter in Stochastic Models." *arXiv*. [arXiv:2510.06439](#). Accepted, Journal of ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part A: Civil Engineering.
3. **Yadav, A.** & Zhang, R. (2025). "Stochastic Subspace via Probabilistic Principal Component Analysis for Characterizing Model Error." *Computational Mechanics*. [doi:10.1007/s00466-025-02701-6](#).

SELECTED CONFERENCE PRESENTATIONS

1. **A. Yadav** & R. Zhang, Stochastic reduced-order modeling for model error characterization and correction, 18th US National Congress on Computational Mechanics (USNCCM), Chicago, USA, July 20-24, 2025.
2. **A. Yadav** & R. Zhang, Stochastic subspace via bootstrap for model-form uncertainty, *Conference on Applied AI & Scientific Machine Learning (CASML)*, IISc, Bangalore, India, December 14-18, 2024.
3. **A. Yadav** & R. Zhang, Stochastic subspace via probabilistic principal component analysis for model-form uncertainty, 16th World Congress on Computational Mechanics & 4th Pan American Congress on Computational Mechanics (WCCM/PANACM), Vancouver, Canada, July 21-26, 2024.

HONORS AND AWARDS

Student Travel Award for WCCM/ECCOMAS 2026	03/2026
Jimmie A. Schindewolf Academic Scholarship , UH	08/2024 - 05/2025
Presidential Fellowship , UH	08/2023 - 05/2025
Finalist , UQ-TTA Student Paper Competition, WCCM/PANACM	07/2024
Cullen Fellowship Travel Grant , UH	05/2024
Top 0.35 % in IIT-JEE (out of 1.4 million candidates)	06/2014

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant , Foundation Engineering (CIVE 4369), UH	01/2026 - 05/2026
Reciter , Mechanics-I Statics (ENGR 2301), UH	08/2024 - 12/2024
Teaching Assistant , Mechanics-I Statics (ENGR 2301), UH	08/2023 - 12/2023

WORK EXPERIENCE

Senior Project Engineer - Indian Oil Corporation Limited	07/2018 - 09/2020
Managed civil execution for an energy-efficient green building and process-unit expansion; coordinated contractors, code compliance, and schedules across multiple work fronts.	

RELEVANT COURSEWORK

Mathematics for AI, Deep Learning, Learning with Data, Data-Driven Engineering, Bayesian Statistics, Optimization Methods, Numerical Methods, Finite Element Method, Structural System Identification.

EXTRA CURRICULAR

Secretary/Joint Secretary , Taekwondo, Institute Sports Council, IIT Roorkee	2016 - 2018
Organized training for 50+ students and coordinated participation in state/national competitions.	
First Dan Black Belt in Taekwondo by World Taekwondo Federation	08/2017
Represented State Uttarakhand in National Taekwondo Championship	12/2017
Organiser and Instructor , Self Defense Camp, Unnat Bharat Abhiyaan	03/2017